# Related work: active inference, federated Bayes, and the scoped bridge

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# Related work (part 2)

This work sits at the boundary between two research communities with different objects of inference. Each cited thread contributes a real component; this paper extends the intersection without claiming that either literature is exhausted. The positioning is therefore against the reviewed sources and their assumptions,

# Related work (part 3)

not against an absolute claim about everything the field has or has not done.

# Pre-modern probability, inverse probability, and collective judgment

These historical sources provide conceptual lineage, not evidence that early authors anticipated KL minimization, active inference, or federation. Pascal and Fermat, Huygens, Montmort, Bernoulli,

# Pre-modern probability... (part 2)

and de Moivre made expectation and uncertain evidence objects of calculation (Pascal and Fermat 1654; Huygens 1657; Montmort 1708; Bernoulli 1713; Moivre 1718).

# Pre-modern probability... (part 3)

Bayes and Price framed inference from observed events to an unknown chance, and Laplace generalized inference about causes (Bayes 1763; Laplace 1774). Their relevance here is vocabulary for generative-model inversion. The executable recovery identity is instead a modern,

# Pre-modern probability... (part 4)

project-local result of sec. 8.8 and sec. 9.

# Pre-modern probability... (part 5)

Daniel Bernoulli's expected-utility treatment and the aggregation work of Borda and Condorcet connect uncertain judgment to action and collective choice (Bernoulli 1738; Borda 1784; Condorcet 1785). They motivate questions about competence, independence, weights, and stakes.

# Pre-modern probability... (part 6)

They do not formally support this paper's log-pool, generalized-Bayes, or robustness claims; the full genealogy remains documented in the source audit.

# Active inference: generative agents, EFE, and colonies

The free-energy principle (Friston 2010) and discrete state-space process theory (Friston et al. 2017) connect generative models, variational inference, and expected-free-energy action selection.

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# Active inference (part 2)

The discrete synthesis (Da Costa et al. 2020), pymdp (Heins et al. 2022), and RxInfer (Bagaev et al. 2023) provide executable machinery.

# Active inference (part 3)

This project reimplements the categorical substrate and its risk–ambiguity decomposition (eq. 22, eq. 23, fig. 11), then evaluates explicit robustness controls at belief fusion.

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# Active inference (part 4)

The fusion operator itself is not new as a mathematical object. Under the explicit categorical posterior-log-potential and fixed-weight bridge of sec. 8.8,

# Active inference (part 5)

the message-combination term of Friston's belief-sharing equation is represented by a logarithmic opinion pool (Genest and Zidek 1986; Genest et al. 1986) and a product-of-experts consensus (Hinton 2002).

# Active inference (part 6)

That limited representation is not a statement that the complete source protocol is a log pool. Abbas' KL view of linear and log-linear pools makes the same point in scoring-rule language: log-pooling can be justified as a KL aggregation rule for expert distributions, not as a contamination-robust estimator (Abbas 2009).

# Active inference (part 7)

The Bayesian Committee Machine adds the distributed-learning analogue: independent estimators trained on data subsets can be combined by a product-style Bayesian rule, with assumptions about conditional independence and prior accounting made explicit (Tresp 2000).

# Active inference (part 8)

Fully Bayesian aggregation gives the social-choice counterpart: geometric pooling of beliefs is singled out by dynamic Bayesian rationality conditions, not by robustness to contaminated reports (Dietrich 2021).

# Active inference (part 9)

That classical literature is useful precisely because it names the hidden commitments — shared support, weight choice, prior accounting, and external-Bayes coherence — that are easy to overlook when the same operation appears as a colony update.

# Active inference (part 10)

A parallel thread studies how active-inference agents coordinate as ensembles — collective behavior and surprise minimization across a colony (Heins et al. 2024), epistemic communities (Albarracin et al. 2022), and collective intelligence (Kaufmann et al. 2021).

# Active inference (part 11)

The belief-sharing thread (Friston et al. 2024) is one member of this family. Agents fuse beliefs, and our colony scenario is a reduced categorical standard- Bayes-limit analogue of its worked example (eq. 14, sec. 10.2).

# Active inference (part 12)

Structure learning completes the conceptual frame. Active inference with Bayesian optimal design selects and reduces models (Smith et al. 2022), while post-hoc BMR (Friston and Penny 2011) motivates our configured sign control (eq. 20, sec. 10.4).

# Active inference (part 13)

Among the active-inference sources reviewed here, belief fusion is not systematically characterized under explicit contamination or intentionally wrong broadcasts, nor connected to the robustness theory used here. Fusion is treated as trusting in that scoped comparison.

# Active inference (part 14)

Friston et al. (2024) present communicating-colony free-energy convergence, Dirichlet language acquisition, and Bayesian model reduction. We use reduced categorical standard-Bayes-limit analogues of those mechanisms (sec. 10.1) before the robust extension.

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# Active inference (part 15)

The third is a configured sign control on a fixed posterior. None is an exact source-protocol or figure reproduction.

# Robust and federated Bayes outside active inference

Federated learning aggregates models trained on decentralized data (McMahan et al. 2017), and the probabilistic-federation line recasts that aggregation as variational inference —

# Robust and federated Bayes... (part 2)

partitioned variational inference (Ashman et al. 2022) and its federated predecessor (Bui et al. 2018). This is a different use of the word *federated* from Friston et al.'s federated inference:

# Robust and federated Bayes... (part 3)

Friston federates hidden-state beliefs among agents sharing a world model, while machine-learning federated learning usually federates parameter or predictive-model updates over decentralized datasets.

# Robust and federated Bayes... (part 4)

The bridge claimed here is therefore algebraic and variational — a shared normalized product/log-pool operator at the KL/NLL recovery corner — not a claim that either source paper already solved the other's problem.

# Robust and federated Bayes... (part 5)

Robustness, meanwhile, has a mature Bayesian theory: general Bayesian updating through a loss (Bissiri et al. 2016), Gibbs posterior inference (Jiang and Tanner 2008), safe learning-rate selection under misspecification (Grünwald 2012),

# Robust and federated Bayes... (part 6)

coarsened posteriors for robustness to exact-data conditioning (Miller and Dunson 2018), divergence-criteria posterior updating (Jewson et al. 2018), Bayesian misspecification asymptotics (Kleijn and Vaart 2012).

# Robust and federated Bayes... (part 7)

The optimization-centric GVI view (Knoblauch et al. 2022) and recent closed-form characterizations (Nguyen and Westerhout 2026) connect objectives to approximating families.

# Robust and federated Bayes... (part 8)

Bounded-influence approaches include density-power and gamma-divergence estimation (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015), robust-divergence variational inference (Futami et al. 2018), and generalized cross-entropy (Zhang and Sabuncu 2018).

# Robust and federated Bayes... (part 9)

Huber and Ronchetti supply the robust-statistics vocabulary of influence and breakdown (Huber and Ronchetti 2009), used here for boundedness and empirical failure modes rather than theorem transfer.

# Robust and federated Bayes... (part 10)

FedGVI (Mildner, Hamelijnck, et al. 2025) supplies the per-agent generalized-Bayes synthesis with client and server divergence choices.

# Robust and federated Bayes... (part 11)

This literature provides decentralized aggregation and theorem-backed results in its stated settings. Among the sources reviewed here, it does not evaluate active-inference POMDP belief consensus—the generative-model-bearing, action-selecting setting where beliefs drive behavior.

# Robust and federated Bayes... (part 12)

The 2025 preprint on convergence rates under prior misspecification (Mildner, Giampouras, et al. 2025) sharpens the current GVI context: bounded divergences can support concentration and rates under explicit assumptions,

# Robust and federated Bayes... (part 13)

but the result does not transfer automatically to this repository's finite categorical state space or to its server-side aggregation heuristic.

# Robust and federated Bayes... (part 14)

The federated-learning robustness literature also supplies important negative space for this paper. Byzantine-tolerant gradient aggregation (Blanchard et al. 2017), geometric-median robust aggregation (Pillutla et al. 2022),

# Robust and federated Bayes... (part 15)

and divergence-weighted gamma-mean aggregation (Li et al. 2022) attack corrupted client updates directly. A recent Bayesian robust-aggregation preprint likewise models unknown client honesty for federated model updates (Karakulev et al. 2025).

# Robust and federated Bayes... (part 16)

Those are close comparators for adversarial federation, but the object being aggregated differs: they aggregate model-update vectors or posterior measures, while this paper attacks categorical belief fusion and generalized-Bayes client updates.

# Robust and federated Bayes... (part 17)

Robust subset-posterior combination is another nearby Bayesian route (Minsker et al. 2017), but it combines posterior measures across data shards rather than active-inference belief broadcasts with a shared latent state.

# Robust and federated Bayes... (part 18)

We use those sources to position the problem, not to import their guarantees into `robust_aggregate`.

# The specific bridge added here

The scoped bridge evaluated by this manuscript is generalized-Bayes belief fusion for active-inference ensembles. Concretely:

# The specific bridge added... (part 2)

**Per-agent bounded-loss updates.**
FedGVI-faithful $\beta$/rcce updates carry the cited bounded-influence result only under its source assumptions (eq. 7, eq. 8).

# The specific bridge added... (part 3)

**Server-side divergence reweighting.** The complementary heuristic has a stated recovery limit (eq. 10). Its finite empirical behavior does not borrow theorems from robust federated aggregation.

# The specific bridge added... (part 4)

**Recovery of the standard-Bayes corner.** The KL/NLL client limits recover Bayes, while the separate zero-robustness server identity returns the project log-linear pool (eq. 11, eq. 6, eq. 10). Under the categorical bridge, that pool specializes only the Eq.

# The specific bridge added… (part 5)

7 message-combination term (Friston et al. 2024).

# The specific bridge added... (part 6)

**Additional executed studies.** The source simulations motivate mechanism analogues, not protocol recovery checks. Added studies cover declared contamination (sec. 10.5), active movement (sec. 17.7), two- and three-level POMDPs (sec. 17.9,

# The specific bridge added... (part 7)

sec. 17.11), finite acuity-by-colony sensitivity, and finite-grid acuity recovery (sec. 17.13, sec. 18).

# The specific bridge added… (part 8)

**Declared paired analysis.** The configured comparison uses matched-pairs Wilcoxon tests (Wilcoxon 1945; Fay and Proschan 2010), Benjamini–Hochberg FDR adjustment (Benjamini and Hochberg 1995), percentile-bootstrap intervals (Efron and Tibshirani 1993),

# The specific bridge added... (part 9)

rank/effect-size caveats (Nakagawa and Cuthill 2007), and an observed-effect planning approximation. These additions are project analyses, not claims about the source paper.

# The specific bridge added… (part 10)

**Objective-backed server control.**
`variational_aggregate` descends its stated free energy (eq. 12); a converged fixed point is coordinatewise stationary. The rule has a raw effective-weight bound and a finite redescending diagnostic (fig. 20).

# The specific bridge added... (part 11)

Whether an equally defensible objective has the sharper reverse-KL heuristic as its minimizer remains open.

# The specific bridge added... (part 12)

The contribution is the bridge: an explicit connection between active-inference belief consensus and robust federated generalized Bayes,

# The specific bridge added... (part 13)

with the standard pool recovered exactly at the corner and the additional studies (the contamination sweep plus several structural extensions) showing how the bridge behaves beyond the Friston et al. baseline.